

8th-Grade

TOPICS

1. Real Numbers
    - Reviews
    - Powers and Radicals
2. Algebraic Expressions
    - Definition of Algebraic Expressions
    - Identities and Factorization
    - Algebraic Ratios
3. Sequences and Functions
    - Sequences
    - Functions
    - Special Functions
4. Equations and Inequalities
    - Equations
    - System of Equations
    - Inequalities
5. Quadratic Equations
6. Probability and Statistics
    - Probability
    - Statistics
7. Mathematical Logic
8. Equations and Inequalities
    - Basic Geometric Concepts
    - Angle and Similarity in Triangles
    - Right Triangles and Trigonometry
    - Quadrilaterals

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2. Algebraic Expressions

Definition of Algebraic Expressions

2. ALGEBRAIC EXPRESSIONS

2.1 Definition of Algebraic Expressions

Definition variable

A **variable** is a letter that is used to represent a numerical quantity. We often use a lower-case letter such as a, b, c, etc. for a variable.

In the example above, x represents a number of years. x is a variable, and 15 + x is called an **algebraic expression**.

Definition algebraic expression

An **algebraic expression** is a combination of numbers, variables, operations and grouping signs.

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2. ALGEBRAIC EXPRESSIONS

2.1 Definition of Algebraic Expressions

EXAMPLE 1.1

Translate each phrase into an algebraic expression.

a. 13 more than a number

b. 5 less than three times a number

c. the square of a number decreased by 7

SOLUTION

a. $13 + x$

b. $3y - 5$

c. $z^2 - 7$

Remember that we can use any letter as a variable, so we could also write $13 + a$, $13 + b$, $13 + c$ etc. or $3p - 5$, $3t - 5$, $3m - 5$ as answers in this example.

2. ALGEBRAIC EXPRESSIONS

2.1 Definition of Algebraic Expressions

EXAMPLE 1.2

Translate each phrase into an algebraic expression.

a. 6 more than the square of a number

b. the quotient of a number squared and 8

c. the product of a number squared and 4

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2. ALGEBRAIC EXPRESSIONS

2.1 Definition of Algebraic Expressions

EXAMPLE 1.3

A restaurant charges 50 cents for one lahmacun and 30 cents for one glass of ayran. Write an algebraic expression for the cost C of a lahmacun and ayran meal.

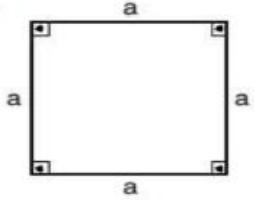
2. ALGEBRAIC EXPRESSIONS

2.1 Definition of Algebraic Expressions

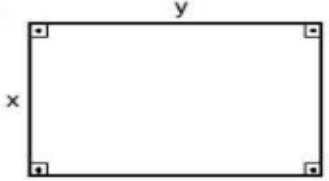
EXAMPLE 1.4

Write an expression for the perimeter of each figure.

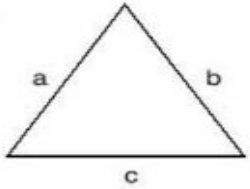
a.



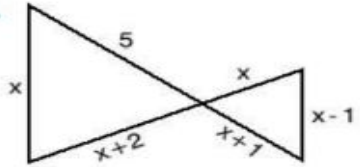
b.



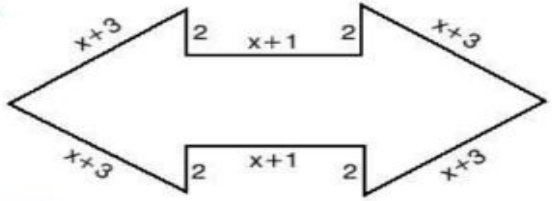
c.



d.



e.



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2. ALGEBRAIC EXPRESSIONS

2.1 Definition of Algebraic Expressions

PRACTICE PROBLEMS

1. Translate each phrase into an algebraic expression.

   a. 13 less than a number

   b. the product of –3 and a number

   c. 15 divided by a number

   d. a number multiplied by 6

   e. the quotient of the square of a number and another number

   f. 5 more than a number, multiplied by 3

   g. 7 times a number, less than 15

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2. ALGEBRAIC EXPRESSIONS

2.1 Definition of Algebraic Expressions

PRACTICE PROBLEMS

2. Match the phrases and the expressions.

a. the quotient of a number squared and 5 more than another number	1. $5 \cdot (x + 4)$
b. the product of 3 less than a number and 12	2. $2 \cdot (x - 3)$
c. five times the sum of 4 and a number	3. $\frac{x^2}{5 + y}$
d. the quotient of a number and the product of 9 and another number	4. $a + \frac{5}{b}$
e. the sum of a number and the quotient of 5 and another number	5. $(x - 3) \cdot 12$
f. twice the difference of a number and 3	6. $\frac{m}{9n}$

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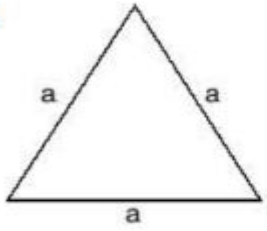
2. ALGEBRAIC EXPRESSIONS

2.1 Definition of Algebraic Expressions

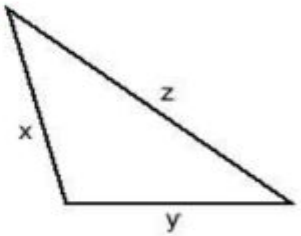
PRACTICE PROBLEMS

3. Write an algebraic expression for the perimeter of each geometric figure.

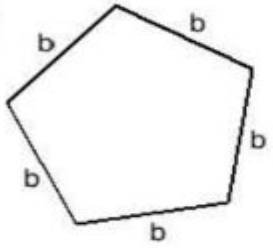
a.



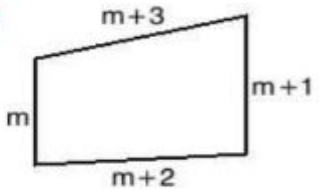
b.



c.



d.



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2. ALGEBRAIC EXPRESSIONS

2.1 Definition of Algebraic Expressions

PRACTICE PROBLEMS

4. Ahmet is x years old. Write an algebraic expression to answer each question.

a. What was Ahmet's age five years ago?

b. Betul is three years older than Ahmet. How old is Betul?

c. How old was Betul seven years ago?